Irabiel Romero

Applied Mathematics Ph.D. (Dec. 2026) | Scientific Computing & Machine Learning

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Summary

Applied mathematician building scalable numerical methods for estimation and optimization under uncertainty: Bayesian inverse problems, optimal sensor placement, large-scale nonlinear and PDE-constrained optimization, multigrid solvers, and ML surrogates inside constrained optimizers. Three research programs at Lawrence Livermore National Laboratory.

Education

Ph.D., Applied Mathematics, University of California, Merced — Expected Dec. 2026

Thesis: Scalable Methods for Estimation of Power Systems under Uncertainty (Advisors: Noemi Petra, Roummel Marcia)

M.S., Applied Mathematics, California State University, Long Beach — May 2021

Thesis: Numerical Methods for Multiscale Modeling in Bacterial Colony Growth (Outstanding Thesis Award)

B.S., Applied Mathematics and **B.A., Physics**, California State University, Long Beach — May 2018

Experience

Graduate Research Intern, CASC, Lawrence Livermore National Laboratory *(Livermore, CA)* — Jun. 2026 – Aug. 2026

- Integrated a machine learning surrogate into a large-scale nonlinear optimization solver, replacing stability constraints that are computationally intractable to enforce directly.
- Designed and ran validation tests comparing surrogate-constrained solutions against full simulation.

Graduate Student Researcher, Lawrence Livermore National Laboratory *(Remote)* — Jan. 2025 – May 2026

- Extended LLNL's security-constrained AC optimal power flow solver (ExaJuGO, Julia) to analyze wildfire-driven multi-failure contingencies on the California Test System; built interactive visualization tools.
- Developed a CNN surrogate for generator stability and embedded it in the optimizer to enforce transient stability constraints that would otherwise be computationally intractable.

Graduate Student Researcher, University of California, Merced *(Merced, CA)* — Jan. 2023 – Present

- Formulated a Bayesian inverse problem to estimate system states and uncertain model parameters from sparse sensor data, and an optimal experimental design method for sensor placement.
- Developed a generic algebraic multigrid solver for M-matrices and a PDE-constrained inverse problem (advection–diffusion) that recovers a diffusion parameter from sparse data.
- Led an LLNL project with CASC and the UQ group (Fall 2026): built an ML surrogate that learns the distribution of stability outcomes over all fault locations to estimate the probability of instability, with conformal prediction giving calibrated, distribution-free bounds on that probability.

Data Science Intern, Lawrence Livermore National Laboratory *(Livermore, CA)* — Summers 2023, 2024

- Data Science Summer Institute (2024): built a parallel framework (C++, MPI, HDF5) using particle swarm optimization to generate large simulation databases; trained CNNs to classify heartbeat irregularities from ECG signals.
- Data Science Challenge (2023): led a team developing a 1D CNN that reconstructs the heart's transmembrane voltages and activation times from ECG signals.

Graduate Researcher, CSU Long Beach & UC San Diego *(Long Beach and San Diego, CA)* — Jun. 2018 – May 2021

- Developed a Newton–multigrid solver for stiff nonlinear reaction–diffusion PDEs, coupled with a kinetic Monte Carlo simulation.

Selected Publications

- **I. Romero** et al. "Wildfire-driven N-k contingency analysis for security-constrained AC optimal power flow in the California power grid." IEEE EPEC, 2026.
- **I. Romero**, R. Marcia, I. Aravena, N. Petra. "Optimal PMU placement for state estimation with grid parameter uncertainty." IEEE PES Grid Edge, 2025.
- A. Ranganath, **I. Romero**, M. Singhal, R. Marcia. "Quasi-Adam: Accelerating Adam using quasi-Newton approximations." IEEE ICMLA, 2024.

Skills & Awards

Languages & HPC:	Python, C++, Julia, MATLAB; MPI, OpenMP, HDF5, Git
Methods:	PDE-constrained optimization, inverse problems, Bayesian inference, UQ, conformal prediction
ML & libraries:	PyTorch, TensorFlow, JAX, SciPy, IPOPT, FEniCS, hIPPYlib; CNNs, surrogate modeling
Awards:	SIAM NCC poster awards (2025, 2026); Fields Family Fellowship; Outstanding Thesis Award